

ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems

Ex.N.3: Spectrum Analysis of Sub-1 GHz, Mid-band & mmWave by Plotting Propagation

Loss. Date

Objective 1: Matlab Code and Output:

Program:

```
clc; clear; close all;
```

```
d = 0.1:0.1:5; % Distance (km)
```

```
f = [900 3500 28000]; % Frequency (MHz)
```

```
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
```

```
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
```

```
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
```

```
figure
```

```
% ----- Graph 1 -----
```

```
subplot(2,1,1)
```

```
plot(d,PL1,'b','LineWidth',2); hold on
```

```
plot(d,PL2,'r','LineWidth',2)
```

```
plot(d,PL3,'k','LineWidth',2)
```

```
title('Propagation Loss vs Distance')
```

```
xlabel('Distance (km)')
```

```
ylabel('Loss (dB)')
```

```
grid on
```

```
% ----- Graph 2 (NO legend used to avoid error) -----
```

```
subplot(2,1,2)
```

```
Freq = [900 3500 28000];
```

```
Loss = [PL1(10) PL2(10) PL3(10)];
```

```
plot(Freq,Loss,'-o','LineWidth',2)
```

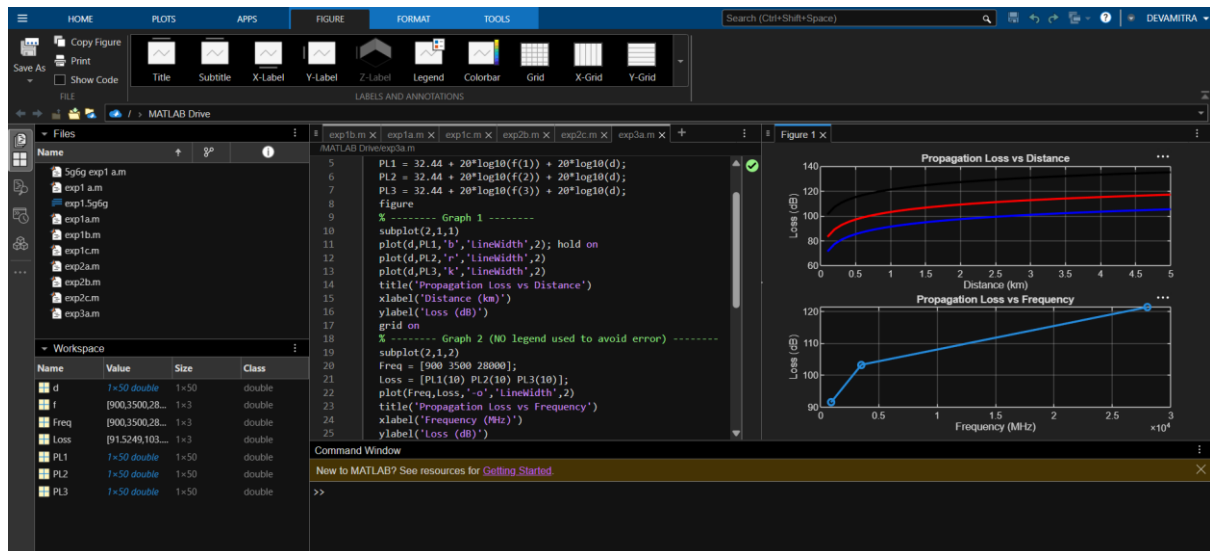
```
title('Propagation Loss vs Frequency')
```

xlabel('Frequency (MHz)')

ylabel('Loss (dB)')

grid on

Output:



Objective 2: Matlab Code and Output:

Program:

clc; clear; close all;

d = 2; % fixed distance (km)

f = [900 3500 28000]; % MHz

PL = 32.44 + 20*log10(f) + 20*log10(d);

figure

% ----- Graph 1: Frequency vs Loss -----

subplot(2,1,1)

plot(f,PL,'-o','LineWidth',2)

title('Propagation Loss vs Operating Frequency')

xlabel('Frequency (MHz)')

ylabel('Propagation Loss (dB)')

grid on

% ----- Graph 2: Band Comparison -----

subplot(2,1,2)

bar(PL)

title('Comparison of Propagation Loss Across Bands')

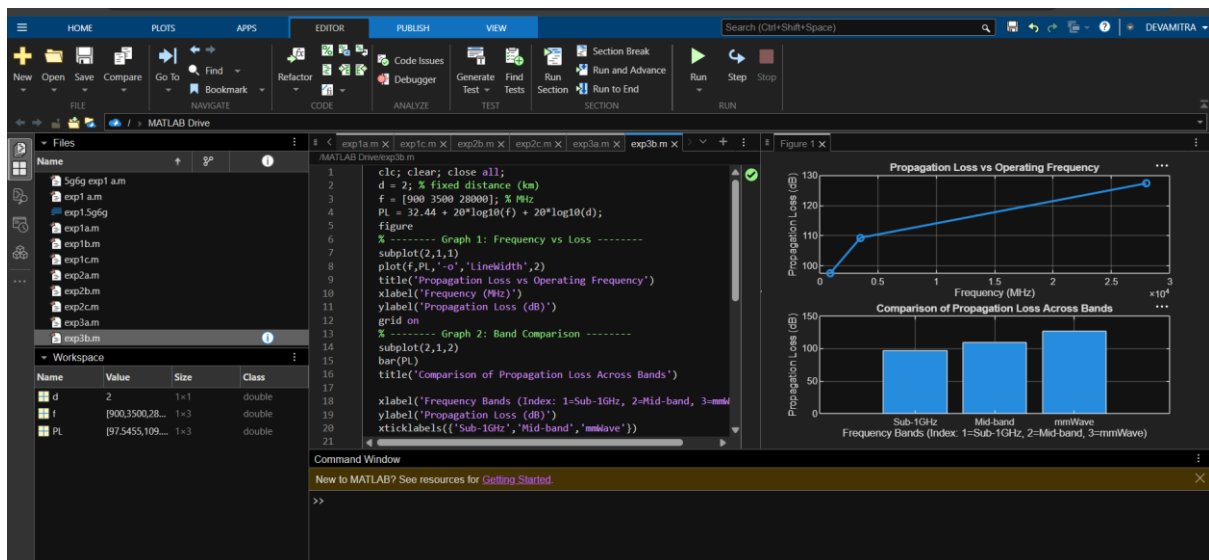
xlabel('Frequency Bands (Index: 1=Sub-1GHz, 2=Mid-band, 3=mmWave)')

ylabel('Propagation Loss (dB)')

xticklabels({'Sub-1GHz','Mid-band','mmWave'})

grid on

Output:



Objective 3: Matlab Code and Output:

Program:

clc; clear; close all;

% Frequency bands (MHz)

f = [900 3500 28000];

% Distance (km)

d = 2;

% Propagation Loss (FSPL)

$$PL = 32.44 + 20 \cdot \log_{10}(f) + 20 \cdot \log_{10}(d);$$

figure

% ----- Graph 1: Simple Line Plot (SAFE) -----

subplot(2,1,1)

plot(f, PL, '-o', 'LineWidth', 2)

title('Propagation Loss vs Operating Frequency')

xlabel('Frequency (MHz)')

ylabel('Propagation Loss (dB)')

grid on

% ----- Graph 2: Safe Comparison Plot -----

subplot(2,1,2)

plot(1:3, PL, '-s', 'LineWidth', 2)

title('Band-wise Propagation Loss Comparison')

xlabel('Band Index (1=Sub1GHz, 2=Mid-band, 3=mmWave)')

ylabel('Propagation Loss (dB)')

grid on

Output:

